

SQL Queries Output (Using DB Browser)

1. Total revenue per category (revenue = quantity × unit_price × (1 - discount_percent/100))

	category	total_revenue
1	Electronics	282471902.7
2	Home	279811251.22
3	Sports	271972272.35
4	Clothing	261534400.03
5	Books	245588391.44

2. Top 10 customers by total order value

	customer_id	customer_name	total_order_value
1	C0526	Rebecca Howard	4645228.58
2	C0456	Dominic Brown	4592392.86
3	C0404	Samantha Jones	3739404.8
4	C0719	Michelle Evans	3603473.95
5	C0594	Laura Jacobs	3598888.92
6	C0198	Michael Vargas	3572756.17
7	C0877	Jordan Jones	3561710.69
8	C0976	Brian Grant	3505681.84
9	C0683	Brett Smith	3295658.73
10	C0904	Jay Fuentes	3290530.46

3. Month-wise order count for the last 12 months

	order_month	total_orders
3	2025-09	185
4	2025-10	242
5	2025-11	183
6	2025-12	221
7	2026-01	208
8	2026-02	188
9	2026-03	219
10	2026-04	214
11	2026-05	215
12	2026-06	215
13	2026-07	75
14	2026-09	2
15	2026-10	5
16	2026-11	2

4. Find customers who placed orders but never had any item delivered

	customer_id	customer_name
1	C0005	Catherine Thompson
2	C0009	Julia Buck
3	C0011	Margaret Ryan
4	C0014	Tiffany Morton
5	C0017	Lindsey Miller
6	C0026	Michele Ruiz
7	C0027	James Flores
8	C0028	Patrick Miller
9	C0029	Steven Patel
10	C0031	Dr. William Grant
11	C0034	Donald Smith
12	C0037	Johnathan Frazier
13	C0041	Nicholas Jimenez
14	C0043	Becky Garrett
15	C0045	Alyssa Oconnor

5. Products that were ordered but had more returns than purchases

	product_id	product_name	returned_quantity	purchased_quantity
1	P0004	Harris, Grant And Pratt Laptop	39	29
2	P0074	Taylor-Romero Novel	29	28
3	P0304	Wise-Perez Comics	32	31

6. Calculate the return rate (returned items / total items) per category

	category	returned_items	total_items	return_rate_percent
1	Books	1324	6187	21.4
2	Sports	1368	6708	20.39
3	Home	1403	6894	20.35
4	Clothing	1340	6698	20.01
5	Electronics	1379	7136	19.32

7. Running Totals with Window Functions

	region_code	order_date	daily_revenue	running_total
1	East	2024-04-09	53449.76	53449.76
2	East	2024-07-11	42411.37	95861.13
3	East	2024-07-12	52222.99	618084.12
4	East	2024-07-14	1082447.02	1700531.14
5	East	2024-07-15	891097.93	2591629.07
6	East	2024-07-16	1201096.11	3792725.18
7	East	2024-07-18	1588098.69	5380823.87
8	East	2024-07-19	1821413.72	7202237.59
9	East	2024-07-20	735479.24	7937716.83
10	East	2024-07-21	52522.83	7990239.66

Execution finished without errors.
Result: 2309 rows returned in 107ms

8. Ranking with DENSE_RANK

	customer_id	customer_name	lifetime_value	customer_rank
1	C0526	Rebecca Howard	4645228.58	1
2	C0456	Dominic Brown	4592392.86	2
3	C0404	Samantha Jones	3739404.8	3
4	C0719	Michelle Evans	3603473.95	4
5	C0594	Laura Jacobs	3598888.92	5
6	C0198	Michael Vargas	3572756.17	6
7	C0877	Jordan Jones	3561710.69	7
8	C0976	Brian Grant	3505681.84	8
9	C0683	Brett Smith	3295658.73	9
10	C0904	Jay Fuentes	3290530.46	10

Execution finished without errors.
Result: 988 rows returned in 36ms

9. LAG/LEAD Analysis

	customer_id	order_date	previous_order_date	days_gap	customer_status
1	C0001	2024-07-14	NULL	NULL	At Risk
2	C0001	2024-08-11	2024-07-14	28.0	At Risk
3	C0001	2024-09-26	2024-08-11	46.0	At Risk
4	C0001	2025-04-21	2024-09-26	207.0	At Risk
5	C0001	2025-10-10	2025-04-21	172.0	At Risk
6	C0001	2026-01-15	2025-10-10	97.0	At Risk
7	C0002	2025-01-14	NULL	NULL	At Risk
8	C0002	2025-05-23	2025-01-14	129.0	At Risk
9	C0002	2025-12-29	2025-05-23	220.0	At Risk
10	C0003	2025-02-23	NULL	NULL	At Risk

Execution finished without errors.
Result: 5000 rows returned in 37ms

10. CTE with Multiple Levels

	revenue_month	spend_category	total_customers
1	2024-02	High	2
2	2024-03	High	3
3	2024-04	High	1
4	2024-06	High	2
5	2024-07	High	107
6	2024-07	Low	3
7	2024-07	Medium	2
8	2024-08	High	172
9	2024-08	Low	5
10	2024-09	High	169

Execution finished without errors.
Result: 72 rows returned in 55ms

11. NTILE for Segmentation

	customer_id	total_value	quartile	quartile_label
1	C0526	4645228.58	1	Platinum
2	C0456	4592392.86	1	Platinum
3	C0404	3739404.8	1	Platinum
4	C0719	3603473.95	1	Platinum
5	C0594	3598888.92	1	Platinum
6	C0198	3572756.17	1	Platinum
7	C0877	3561710.69	1	Platinum
8	C0976	3505681.84	1	Platinum
9	C0683	3295658.73	1	Platinum
10	C0904	3290530.46	1	Platinum

Execution finished without errors.
Result: 988 rows returned in 47ms

12. Year-over-Year Comparison

	year	month	revenue	prev_year_revenue	yoy_growth_percent
1	2025	01	54369178.43	NULL	NULL
2	2026	01	51799437.54	54369178.43	-4.73
3	2024	02	770025.63	NULL	NULL
4	2025	02	52873367.23	770025.63	6766.44
5	2026	02	45796357.42	52873367.23	-13.38
6	2024	03	917448.22	NULL	NULL
7	2025	03	51590235.19	917448.22	5523.23
8	2026	03	56792648.58	51590235.19	10.08
9	2024	04	53449.76	NULL	NULL
10	2025	04	54401897.39	53449.76	101681.37

Execution finished without errors.
Result: 32 rows returned in 53ms

13. First/Last Value Analysis

	customer_id	first_category	last_category	category_shift
1	C0001	Books	Electronics	Yes
2	C0002	Clothing	Home	Yes
3	C0003	Clothing	Clothing	No
4	C0004	Sports	Home	Yes
5	C0005	Books	Home	Yes
6	C0006	Clothing	Sports	Yes
7	C0007	Sports	Home	Yes
8	C0008	Home	Home	No
9	C0009	Home	Electronics	Yes
10	C0010	Sports	Home	Yes

Execution finished without errors.
Result: 989 rows returned in 100ms

14. Cumulative Distribution

	customer_id	revenue	cumulative_revenue	cumulative_percent
1	C0526	4645228.58	4645228.58	0.37
2	C0456	4592392.86	9237621.44	0.73
3	C0404	3739404.8	12977026.24	1.02
4	C0719	3603473.95	16580500.19	1.31
5	C0594	3598888.92	20179389.11	1.59
6	C0198	3572756.17	23752145.28	1.88
7	C0877	3561710.69	27313855.97	2.16
8	C0976	3505681.84	30819537.81	2.43
9	C0683	3295658.73	34115196.54	2.69
10	C0904	3290530.46	37405727.0	2.95

Execution finished without errors.
Result: 988 rows returned in 34ms

15. Complex CTE: Cohort Analysis

	cohort_month	month_0	month_1	month_2	month_3	retention_rate
7	2024-01	0	0	0	0	0
8	2024-02	0	0	0	0	0
9	2024-03	0	1	0	0	0
10	2024-04	0	0	0	4	0
11	2024-05	0	0	5	9	0
12	2024-06	0	1	2	8	0
13	2024-07	3	4	5	5	133.33
14	2024-08	6	5	9	2	83.33
15	2024-09	4	5	7	6	125.0
16	2024-10	3	6	6	9	200.0

Execution finished without errors.
Result: 37 rows returned in 61ms

16. Self-Join with Window Function

	product_a	product_b	times_bought_together
1	P0323	P0463	5
2	P0045	P0264	4
3	P0010	P0120	3
4	P0015	P0432	3
5	P0027	P0038	3
6	P0028	P0357	3
7	P0029	P0083	3
8	P0038	P0102	3
9	P0039	P0201	3
10	P0042	P0176	3

Execution finished without errors.
Result: 13335 rows returned in 84ms

Note – This Pdf Only has screenshots and question for queries please check the aggregation.sql, cohort_analysis.sql and window_function.sql I have only attached the results of queries due to space issue.